

1. $a \mid bc$ $\therefore$ 存在整数 m, n 使 $a = mn$, $m \mid b$, $n \mid c$

$$\therefore m \mid \gcd(a, b) \quad n \mid \gcd(a, c)$$

$$\therefore a = mn \mid \gcd(a, b) \gcd(a, c)$$

2. $x = x_0 + kx_1, \quad y = y_0 + ky_1 \quad (x_1 = \frac{b}{\gcd(a, b)}, \quad y_1 = -\frac{a}{\gcd(a, b)}) \quad k \in \mathbb{Z}$

下证上述解为 $ax + by = c$ 的全部解

充分性: 代入得 $ax_0 + by_0 + akx_1 + bky_1 = c + \frac{kab}{\gcd(a, b)} - \frac{kab}{\gcd(a, b)} = c$

必要性: 对 $ax + by = c$ 有 $a(x - x_0) + b(y - y_0) = 0$

$$\therefore a(x - x_0) + b(y - y_0) = 0$$

$$\therefore \frac{x - x_0}{y_1} + \frac{y - y_0}{x_1} = 0$$

从而有 $x = x_0 + \frac{bk}{\gcd(a, b)}, \quad y = y_0 + \frac{ak}{\gcd(a, b)}$

3. 设 $a = p^{a_0} p_1^{a_1} \dots p_n^{a_n}$, $b = p^{b_0} p_1^{b_1} \dots p_n^{b_n}$ ($a_i, b_i \geq 0$, $b_i + a_i > 0$, $1 \leq i \leq n$) p 为素数

(1) 设 $c_i = \min(a_i, b_i)$ 则 $c = p^{c_0} p_1^{c_1} \dots p_n^{c_n}$

$$\therefore c = \gcd(a, b) \quad \therefore v_p(\gcd(a, b)) = v_p(c) = c_0 = \min\{a_0, b_0\} = \min\{v_p(a), v_p(b)\}$$

(2) 设 $d_i = \max(a_i, b_i)$ 则 $d = p^{d_0} p_1^{d_1} \dots p_n^{d_n}$

$$\therefore d = [a, b] \quad \therefore v_p(\gcd([a, b])) = v_p(d) = d_0 = \max(a_0, b_0) = \max\{v_p(a), v_p(b)\}$$

(3) 设 $c_0 = \min(a_0, b_0)$

$$\therefore a + b = p^{a_0} p_1^{a_1} \dots p_n^{a_n} + p^{b_0} p_1^{b_1} \dots p_n^{b_n} = p^{c_0} (p^{a_0 - c_0} p_1^{a_1} \dots p_n^{a_n} + p^{b_0 - c_0} p_1^{b_1} \dots p_n^{b_n})$$

$$\therefore p^{c_0} \mid a + b \quad \therefore v_p(a + b) \geq c = \min\{v_p(a), v_p(b)\}$$

当 $v_p(a) \neq v_p(b)$ 时, 不妨设 $v_p(a) > v_p(b)$ $a = p^{a_0} a_1, \quad b = p^{b_0} b_1$

$$\therefore a + b = p^{b_0} (p^{a_0 - b_0} a_1 + b_1)$$

$$\therefore p \mid p^{a_0 - b_0} a_1, \quad p \nmid b_1 \quad \therefore p \nmid (p^{a_0 - b_0} a_1 + b_1)$$

$$\therefore p^{b_0} \mid a + b, \quad p^{b_0 + 1} \nmid a + b \quad \therefore v_p(a + b) = \min(v_p(a), v_p(b))$$

当 $v_p(a) = v_p(b)$ 时, 等号成立 $\Leftrightarrow p \nmid a + b$, 即 $p \nmid \frac{a}{p^{v_p(a)}} + \frac{b}{p^{v_p(b)}}$

4. (1) $1, 2, \dots, n$ 中有唯一一项 m , 使其 $v_2(x)$ 最大 设 $v_2(m) = n$

将 $1, 2, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{n}$ 中除 m 外的项与所通分, 其分子为偶数

$\therefore$ 将 $\frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{n}$ 通分, 其分子除 1 为奇数外均为偶数

故通分后加和, 分子为奇数, 分母为偶数, 故 $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{n}$ 不为整数

(2) 若 $a, a+1, \dots, a+n$ 中不存在唯一一项的 $v_2(x)$ 最大, 则取 2 个最大项 m_1, m_2

$$\text{设 } m_1 = v_2(m_1) = v_2(m_2) = n' \quad \therefore m_1 = 2^{n'} d_1, \quad m_2 = 2^{n'} d_2 \quad \therefore d_1, d_2 \text{ 为奇数}$$

$$\therefore \frac{m_1 + m_2}{2} \text{ 在 } m_1, m_2 \text{ 在 } a, a+1, \dots, a+n \text{ 中, } \therefore \frac{m_1 + m_2}{2} \text{ 也在 } a, a+1, \dots, a+n \text{ 中}$$

$$\therefore \frac{m_1 + m_2}{2} = 2^{n'} \frac{d_1 + d_2}{2} \quad \text{若 } \frac{d_1 + d_2}{2} \text{ 为偶数, 则矛盾}$$

若 $\frac{d_1+d_2}{2}$ 为奇数, 则再次取 m_1 与 $\frac{m_1+m_2}{2}$ 中间值,

重复上述操作过次数超过

$\therefore d = \gcd(d_1, d_2)$ 为奇数, 又 $\because m_1, m_2$ 在 $a, a+1, \dots, a+n$ 中 (不妨设 $m_1 < m_2$)

$\therefore \frac{m_1+m_2}{2} (d_1+d_2)$ 在 $a, a+1, \dots, a+n$ 中

而 $\sum V_p(2^n(d_1+d_2)) = n' + \sum V_p(d_1+d_2) \geq n'+1 > n'$ 矛盾

$\therefore a, a+1, \dots, a+n$ 中存在唯一项 $V_2(x)$ 最大, 后续操作同 (1)

$\therefore$ 由勒让德定理, $V_p(n!) = \frac{n - S_p(n)}{p-1} \leq \frac{n-1}{p-1}$

$\therefore$ 设 V_p $n! \leq p^{\frac{n-1}{p-1}}$ 设 $n! = p_1^{q_1} p_2^{q_2} \dots p_k^{q_k}$ ($2 \leq q_i \leq n$), 且 p_i 均为质数)

$\therefore q_i \leq \frac{n-1}{p_i-1}$

$\therefore n! \leq p_1^{q_1} \dots p_k^{q_k} \leq p_1^{\frac{n-1}{p_1-1}} \dots p_k^{\frac{n-1}{p_k-1}} \leq p_1^{\frac{n-1}{p_1-1}} p_k^{\frac{n-1}{p_k-1}}$

$\therefore \sqrt[n]{n!} \leq \prod_{p \leq n} p^{\frac{1}{p-1}}$

若素数有限, 则 $\exists N$, 使 $\forall p, p < N$

取 $N < n \rightarrow \infty$, 则 $\sqrt[n]{n!} \rightarrow \infty$ $\therefore \sqrt[n]{n!} > \prod_{p \leq N} p^{\frac{1}{p-1}} = \prod_{p \leq n} p^{\frac{1}{p-1}}$ 矛盾

$\therefore$ 素数有无穷多个

设 p 数列 $\{p_i\}_{i=1}^{\infty}$ 为从小到大的素数

则 p_i 无平方因子 $\prod_{i=1}^{\infty}$

$\therefore \prod_{i=1}^{\infty} (1 + \frac{1}{p_i}) = \prod_{i=1}^{\infty} (1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \dots) = \prod_{i=1}^{\infty} \frac{1}{1 - \frac{1}{p_i}} = \sum_{n=1}^{\infty} \frac{1}{n} \leq \sum_{n=1}^{\infty} e^{\frac{1}{p_n}}$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n} = \prod_{i=1}^{\infty} (1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \dots) = \prod_{i=1}^{\infty} \frac{1}{1 - \frac{1}{p_i}} \leq \prod_{i=1}^{\infty} e^{\frac{1}{p_i}}$

$\therefore \sum_{n=1}^{\infty} \frac{1}{p_n} \geq \frac{1}{2} \ln(\sum_{n=1}^{\infty} \frac{1}{n}) \rightarrow \infty$

$\therefore \sum_{n=1}^{\infty} \frac{1}{p_n}$ 发散 $\therefore \sum_{n \text{ 无平方因子}} \frac{1}{n}$ 发散

$\therefore \prod_{n=1}^{\infty} (1 + \frac{1}{p_n}) = \sum_{n \text{ 无平方因子}} \frac{1}{n} + 1$ 发散